Lygtys su keitiniais

Lygis: Beast

30 mišrių uždavinių

Užduotis. Pertvarkykite lygtį, įveskite tinkamą keitinį t, išspręskite gautą pilnąją kvadratinę lygtį ir raskite pradinės lygties realiuosius sprendinius. Atsižvelkite į pradinių reiškinių apibrėžimo sritį ir galimas keitinio reikšmes.

1. $x^2(x-3)^2 - 2x^2 + 6x - 8 = 0$

2. $(x^2 - x + 2)^2 = 2x(x-1) + 12$

3. $3^{x+1} + 3^{1-x} = 10$

4. $x^2(x+2)^2 - 5x^2 - 10x + 6 = 0$

5. $(x^2 + 3x + 1)^2 = (x+2)(x+1) + 19$

6. $2^{x+1} + 2^{1-x} = 5$

7. $\dfrac{2x+6}{x-1} + \dfrac{6x-6}{2x+6} = 7$

8. $(x^2 + 2x - 1)^2 = 3(x+1)^2 + 4$

9. $4x^2(x+1)^2 - 6x^2 - 6x + 2 = 0$

10. $5^{x+1} + 5^{2-x} = 30$

11. $\dfrac{3x-6}{x+1} + \dfrac{2x+2}{x-2} = 7$

12. $(x^2+2)^2 - 4x(x^2+2) + 4x^2 - 7x^2 + 14x - 4 = 0$

13. $(2x^2 + 3x + 1)^2 = (2x+1)(x+1) + 6$

14. $2^{x+2} + 2^{3-x} = 18$

15. $\dfrac{x^2+3x+2}{x^2-1} - \dfrac{6x-6}{x+2} = 1$

16. $(x^2+x)^2 + 2(x^2+x)(x-1) + (x-1)^2 - 5x^2 - 10x + 11 = 0$

17. $(x-1)^2(x+2)^2 - 2x^2 - 2x - 4 = 0$

18. $3^{x+1} - 3^{2-x} = 26$

19. $(x^2+2x)^2 + 2(x^2+2x)(x^2-x+1) + (x^2-x+1)^2 - 10x^2 - 5x - 1 = 0$

20. $\dfrac{2x^2+2}{x^2-1} - \dfrac{3x^2-3}{x^2+1} = 1$

21. $(x+1)^2(x-2)^2 - 3x^2 + 3x + 2 = 0$

22. $(2x^2+x+1)^2 - 2(2x^2+x+1)(x^2+3x) + (x^2+3x)^2 - 5x^2 + 10x - 1 = 0$

23. $(x^2+x-3)^2 + x(x+1) = 9$

24. $\dfrac{x^2-x-2}{x^2-4} - \dfrac{2x+4}{x+1} = 1$

25. $(x^2 - 4x + 5)^2 = 2(x-1)(x-3) + 19$

26. $2x^2(2x-1)^2 - 6x^2 + 3x + 1 = 0$

27. $(x^2-1)^2+(x+1)^2+2(x^2-1)(x+1)-8x^2-8x+12=0$

28. $\left(\frac{1}{2}\right)^{x-1}+\left(\frac{1}{2}\right)^{-x}=3$

29. $\frac{4x^2+4x+4}{2x^2+2x-2}+\frac{3x^2+3x-3}{x^2+x+1}=7$

30. $(x^2+x+2)^2+(x^2-2x+1)^2-2(x^2+x+2)(x^2-2x+1)-3x-7=0$

Atsakymai

1. $x_1 = -1, \quad x_2 = 1, \quad x_3 = 2, \quad x_4 = 4$

2. $x_1 = -1, \quad x_2 = 2$

3. $x_1 = -1, \quad x_2 = 1$

4. $x_1 = -3, \quad x_2 = -1 - \sqrt{3}, \quad x_3 = -1 + \sqrt{3}, \quad x_4 = 1$

5. $x_1 = -4, \quad x_2 = 1$

6. $x_1 = -1, \quad x_2 = 1$

7. $x_1 = -7, \quad x_2 = 3$

8. $x_1 = -1 - \sqrt{7}, \quad x_2 = -1, \quad x_3 = -1 + \sqrt{7}$

9. $x_1 = \dfrac{-1 - \sqrt{5}}{2}, \quad x_2 = \dfrac{-1 - \sqrt{3}}{2}, \quad x_3 = \dfrac{-1 + \sqrt{3}}{2}, \quad x_4 = \dfrac{-1 + \sqrt{5}}{2}$

10. $x_1 = 0, \quad x_2 = 1$

11. $x_1 = -4, \quad x_2 = \dfrac{7}{2}$

12. $x_1 = -1, \quad x_2 = 0, \quad x_3 = 2, \quad x_4 = 3$

13. $x_1 = -2, \quad x_2 = \dfrac{1}{2}$

14. $x_1 = -1, \quad x_2 = 2$

15. $x_1 = 0, \quad x_2 = \dfrac{5}{2}$

16. $x_1 = -1 - \sqrt{5}, \quad x_2 = -3, \quad x_3 = 1, \quad x_4 = -1 + \sqrt{5}$

17. $x_1 = -3, \quad x_2 = -1, \quad x_3 = 0, \quad x_4 = 2$

18. $x_1 = 2$

19. $x_1 = -\dfrac{3}{2}, \quad x_2 = -\dfrac{1}{2}, \quad x_3 = 0, \quad x_4 = 1$

20. $x_1 = -\sqrt{5}, \quad x_2 = 0, \quad x_3 = \sqrt{5}$

21. $x_1 = -2, \quad x_2 = \dfrac{1 - \sqrt{5}}{2}, \quad x_3 = \dfrac{1 + \sqrt{5}}{2}, \quad x_4 = 3$

22. $x_1 = -1, \quad x_2 = 0, \quad x_3 = 2, \quad x_4 = 3$

23. $x_1 = \dfrac{-1 - \sqrt{21}}{2}, \quad x_2 = -1, \quad x_3 = 0, \quad x_4 = \dfrac{-1 + \sqrt{21}}{2}$

24. $x_1 = -3, \quad x_2 = -\dfrac{3}{2}$

25. $x_1 = 0, \quad x_2 = 4$

26. $x_1 = -\dfrac{1}{2}, \quad x_2 = \dfrac{1 - \sqrt{5}}{4}, \quad x_3 = \dfrac{1 + \sqrt{5}}{4}, \quad x_4 = 1$

27. $x_1 = -3, \quad x_2 = -2, \quad x_3 = 1, \quad x_4 = 2$

28. $x_1 = 0, \quad x_2 = 1$

29. $x_1 = -2, \quad x_2 = 1$

30. $x_1 = -1, \quad x_2 = \dfrac{2}{3}$

Keitinių užuominos

1. $t = x(x-3)$

2. $t = x^2 - x + 2$

3. $t = 3^x$

4. $t = x(x+2)$

5. $t = x^2 + 3x + 1$

6. $t = 2^x$

7. $t = \dfrac{x+3}{x-1}$

8. $t = x^2 + 2x - 1$

9. $t = 2x(x+1)$

10. $t = 5^x$

11. $t = \dfrac{x-2}{x+1}$

12. $t = x^2 - 2x + 2$

13. $t = 2x^2 + 3x + 1$

14. $t = 2^x$

15. $t = \dfrac{x+2}{x-1}$

16. $t = x^2 + 2x - 1$

17. $t = (x-1)(x+2)$

18. $t = 3^x$

19. $t = 2x^2 + x + 1$

20. $t = \dfrac{x^2+1}{x^2-1}$

21. $t = (x+1)(x-2)$

22. $t = x^2 - 2x + 1$

23. $t = x^2 + x - 3$

24. $t = \dfrac{x+1}{x+2}$

25. $t = x^2 - 4x + 5$

26. $t = x(2x-1)$

27. $t = x^2 + x$

28. $t = \left(\dfrac{1}{2}\right)^x$

29. $t = \dfrac{x^2+x+1}{x^2+x-1}$

30. $t = 3x + 1$